EE4091 - Photonics

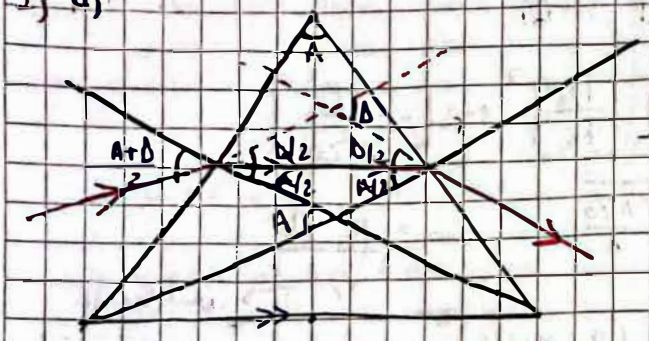
Assignment 1

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Part 1. : Ray Optics

1) a)



$$\frac{\sin\alpha_i}{\sin\alpha_o} = \frac{n_{prism}}{n_o} \qquad \alpha_i = \frac{A+D}{2} \qquad \alpha_o = \frac{A}{2}$$

$$\frac{\sin\left[(A+D)/2\right]}{\sin\left(A/2\right)} = \frac{n_{prism}}{n_o}$$

b)

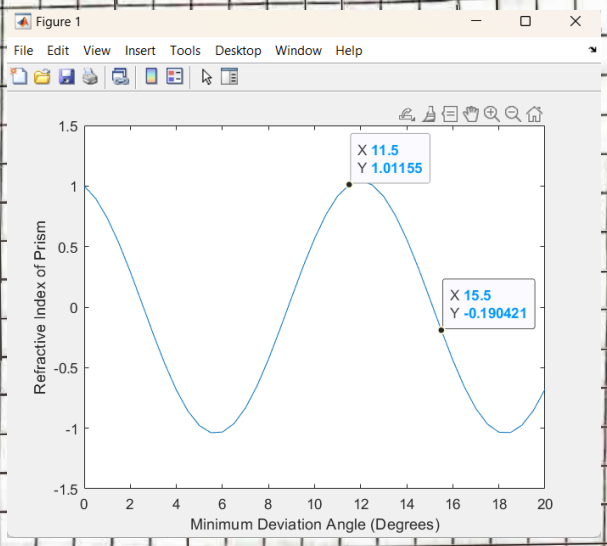
```
A = 10;
d_min = [0:0.5:20];

% no=1

np = sin((A+d_min)/2)/sin(A/2);

plot(d_min, np);

xlabel('Minimum Deviation Angle (Degrees)');
ylabel('Refractive Index of Prism');
```



c) Air has a refractive index of 1.0003. So when the deviation angle is 11.5, it may be air.

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Part II : Beam Optics:

2) Gaussian Beam : $W(z) = W_0 \sqrt{1 + \left(\frac{z}{z_0}\right)^2}$

$$I(\rho, z) = I_0 \left[\frac{W_0}{W_z}\right]^2 \exp\left[-\frac{2\rho^2}{W^2(z)}\right]$$

$W(z)$: beam radius at distance z

W_0 : min beam radius

$I(\rho, z)$: intensity at distance z

I_0 : peak intensity

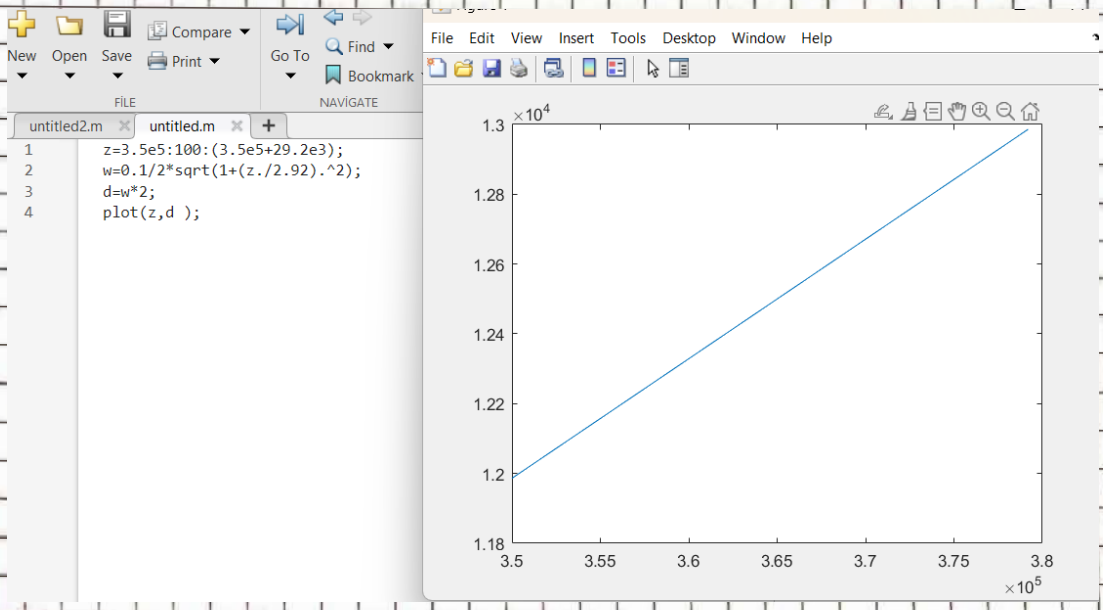
$$W_0 = \sqrt{\frac{\lambda z_0}{\pi}} \qquad z_0 = \frac{\pi \cdot W_0^2}{\lambda}$$

$\lambda \rightarrow$ wavelength

$$z_0 = \frac{\pi \cdot (0.1\,mm/2)^2}{532\,nm} = 2.92\,km$$

The distance to the moon is 3.5×10^5 km

beam diameter for $z = 3.5 \times 10^5$ km to $3.5 \times 10^5 + \overbrace{10 \times z_0}^{29.2}$



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Part III : Wave Optics

3) M identical plane sources with amplitude A and phase shift δ between each configurous source.

$$E = A e^{j\omega t}\left[e^{j\phi_1} + e^{j(\phi_1+\delta)} + \dots + e^{j(\phi_1+(M-1)\delta)}\right]$$

ϕ_1 = phase at the first source ω : angular frequency

general formula

$$1 + r + r^2 + \dots + r^{M-1} = \frac{r^{M-1}}{r-1}$$

→ we will use this formula to simplify the equation.

$$E = A e^{j\omega t}\left[e^{j\phi_1}\frac{e^{jM\delta}-1}{e^{j\delta}-1}\right]$$

using Euler formula:

$$e^{jM\delta} = \cos(M\delta) + j\sin(M\delta)$$

$$E = A e^{j\omega t}\left[e^{j\phi_1}\frac{\cos(M\delta)+j\sin(M\delta)}{\cos(\delta)-j\sin(\delta)}\right]$$

Average intensity:

T: period of wave

$$\langle I\rangle = \frac{1}{T}\int [E(t)]^2\, dt$$

$$\langle I\rangle = \left(\frac{1}{T}\right)\int |A|^2 \left| \left[e^{j\phi_1}\frac{\cos(M\delta)+j\sin(M\delta)}{\cos(\delta)-j\sin(\delta)}\right]\right|^2 dt$$

$$\langle I\rangle = \left(\frac{|A|^2}{T}\right)\frac{M}{2}$$

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